

Dynamic Task Offloading for Multi-UAVs in Vehicular Edge Computing With Delay Guarantees: A Consensus ADMM-Based Optimization

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Abstract—Within the paradigm of forthcoming 6G network infrastructures, unmanned aerial vehicles (UAVs), functioning as principal conveyances, are projected to emerge as pivotal enablers in the nascent domain of the low-altitude economy. UAVs are poised to embrace various innovative applications, including latency-sensitive and compute-intensive services. However, UAVs are constrained by their energy capacity and computational resources, rendering them insufficient for fulfilling the increasingly rigorous service demands in the future. To address these challenges, our investigation focuses on the innovative **UAV-based Vehicular Edge Computing (UVEC) framework**, incorporating Vehicular Edge Computing (VEC) in UAV systems to bolster service reliability. A UAV can enhance its mission duration by dynamically selecting suitable vehicles for computation offloading and adaptively adjusting the task offloading ratio between vehicles and the edge server. By integrating vehicle selection and task offloading scheduling in the UVEC framework, we investigate the optimization of energy efficiency while satisfying the statistical delay and the buffer constraints for UAVs. To deal with the proposed problem, a distributed algorithm is designed by jointly considering the vehicle selection for task offloading radio to vehicles and the edge server. **The stochastic network calculus (SNC)** is employed to derive performance bounds for the statistical delay and constraints, enabling robust analysis and optimization of network performance. After that, we leverage linear transformation techniques to reformulate the original problem into a linear framework, enabling the application of the Alternating Direction Method of Multipliers (ADMM) algorithm to efficiently solve the transformed problem. Theoretical analysis and simulation results show that our algorithm converges while effectively satisfying service reliability constraints within the desired targets, outperforming benchmark schemes in terms of efficiency while meeting task delay and error-rate bounded constraints.

Index Terms—6G, ADMM, stochastic network calculus, unmanned aerial vehicles, vehicular edge computing.

I. INTRODUCTION

WITH the advancement of 6G networks, unmanned aerial vehicles (UAVs), playing the core role in the low-altitude economy, is poised to emerge as a leading-edge technology in future advancements. In UAV-based systems, various innovative applications such as automated patrolling [1] and traffic monitoring [2] are set to rise as key services. The low-altitude economic scenario supported by UAV-based systems combined with 6G **massive ultra-reliable and low-latency communications (mURLLC)** has become an innovative economic paradigm and greatly enriched the convenience of people's lives. In UAV-based scenarios, numerous UAVs are involved in communication, which poses a great challenge to **the high reliability and low latency** in mURLLC [3]. The end-to-end task completion latency is extremely low (<1ms), and the task reliability is extremely high (>99.99999%) in mURLLC scenarios. What's more, energy efficiency is also a key demand for 6G mURLLC systems [4], [5].

To guarantee the service requirements in UAV-based systems, task offloading from UAVs to computing nodes has been proposed as a key enabler to enhance endurance capacity and accelerate computational speed [6]. Nevertheless, the adaptable flight paths of UAVs render a single, stationary edge computing service node insufficient for meeting their diverse requirements. Inspired by vehicles' inherent computing capabilities and ubiquitous nature, vehicular edge computing (VEC) has come up to integrate with UAV systems to support latency-sensitive services with extra-high energy efficiency under the constraint of mURLLC [7]. UAV-based Vehicular Edge Computing (UVEC) systems can meet the mURLLC requirements in 6G networks [8]. Besides, UVEC can effectively handle the problem where the road side unit (RSU) is overloaded [9]. In this novel UVEC framework, each UAV can offload its tasks to nearby vehicles for processing and obtain the results through wireless transmission.

However, ensuring service quality for multi-UAVs in UVEC still encounters numerous challenges. The primary challenge stems from the dynamic mobility of vehicles, resulting in fluctuating network topologies and variable wireless channel

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conditions [10]. Another problem can be caused by **heterogeneous resource limitations** between vehicles and UAVs [11]. Although UAVs can offload computation-intensive tasks to vehicles through wireless networks, task transmission between multiple UAVs and a vehicle can become congested when an excessive number of UAVs converge on the same vehicle for computation. Meanwhile, the computing process in the vehicle could also be blocked due to bounded computing performance in each car. Additionally, the unpredictable nature of task generation in each UAV complicates forecasting the load in UVEC, consequently hindering the achievement of deterministic service constraints. Innovative services in UAV-based systems are delay-sensitive, unlike traditional traffic-heavy applications [12]. As a result, **traditional average metrics are unsuitable for formulating service requirements**.

To address the challenges mentioned above, we use stochastic network calculus (SNC) [13] to describe the stochastic QoS performance of multiple UAVs in UVEC in terms of latency and reliability. Based on the SNC theory, the maximum throughput with buffer constraint and task processing delay violation probability are analyzed in supporting mURLLC. Under the analysis of service reliability for multi-UAVs in UVEC, we concentrate on **maximizing the UAV's energy efficiency problem**. **The energy efficiency of the UAV is the ratio of profit from offloading data to energy consumption**. Based on our system model, we design one distributed algorithm by integrating the vehicle selection and dynamically adjusting task throughput between UAVs and different computation nodes. Finally, to evaluate our proposed algorithm's performance, a set of simulations is performed. The contributions of this article are listed as follows:

- We investigate the problem of maximizing energy efficiency in the UVEC framework. To meet the more stringent performance requirements of UVEC's novel delay-and-reliability-bounded constrained services, we model service reliability using stochastic network calculus.
- The energy efficiency maximization problem with multiple constraints for multi-UAVs in UVEC is highly complex. To solve this issue, we first performed a linear transformation of the problem. After converting it into a linear form, we employed the Consensus-ADMM algorithm to solve the original problem through vehicle selection and dynamic task throughput adjustment.
- Through rigorous theoretical analysis and numerical evaluation, we demonstrate the advantages of our proposed optimization method. Our algorithm achieves convergence and higher energy efficiency while effectively meeting service reliability constraints within the desired targets.

The remainder of the article is organized as follows. Section II summarizes related works. Section III establishes the system model based on UVEC. Section IV analyzes the maximum throughput with buffer constraint and task processing delay violation probability. The energy efficiency maximizing problem is also formulated in this section. Section V proposes a joint task offloading decision and offloading ratio algorithm to address the formulated problem. We conduct simulations to show the

advantage of our algorithm in Section VI. Section VII gives the conclusion.

II. RELATED WORK

Due to the high flexibility and extensive communication coverage of UAVs, incorporating UAVs for task offloading has become a prominent research trend. In UAV-based systems, the UAV's placement is flexible regardless of the usage. The UAV can be positioned anywhere within the scene, rather than being fixed in one location [14]. The mobility of UAVs is invaluable in emergencies, including monitoring natural disaster search and rescue operations [15]. By instantly collecting various types of data such as images, videos, and sound, UAVs can process tasks in real-time by task offloading to an edge cloud nearby [16]. However, the high-speed mobility of UAVs causes fluctuating network topologies and variable wireless channel conditions [17]. In this case, only one fixed and stationary edge server could not support innovative applications in UVECs, such as highway monitoring. To address this issue, the framework of UVEC has been proposed, which leverages multiple UAVs in UVEC. UAVs can offload computation-heavy tasks to the vehicle within the communication range.

However, several critical issues still need to be addressed when deploying sensitive applications in UAVs for the framework of UVEC. The limitations of computing resources in vehicles and communication resources between wireless links are two significant factors constraining service performance in UVEC. If numerous UAVs offload tasks to the same node simultaneously, the task queue at the node may become congested [18]. In addition, the transmission delays and queuing delays of the services are unpredictable due to the random nature of the transmission environment, which can hardly meet the stringent requirements of delay-sensitive services [19]. Furthermore, delay-sensitive services in UVEC necessitate more rigorous service guarantees compared to conventional high-traffic applications [12]. Merely incorporating average metrics into the system model falls short of fulfilling the QoS requisites for these innovative services.

To fulfill the QoS requirements for delay-sensitive services, the congestion in task transmission and edge computing process needs to be addressed. Therefore, various resource scheduling techniques have been recently developed and introduced to meet the stringent delay requirements. Hou et al. [20] proposed an Edge-assisted Adaptive Video streaming system (EAVS) with Serverless pipelines, which facilitates fine-grained management for multiple concurrent video transmission pipelines. Ng et al. [21] proposed a wireless-powered hybrid coding edge computing network in which UAVs act as mobile charging stations to reduce the QoS of energy-constrained IoT device applications. Xu et al. [22] studied the placement of UAVs to collect latency-sensitive data from IoTs. Lin et al. [23] proposed a strategy for multi-UAV enabled communication systems to achieve an optimal trade-off between the global energy efficiency and communication capacity, in which multiple UAV-GT pairs cases are also considered. However, these studies mainly considered task

offloading for nodes with fixed locations. Vehicles in UVEC are mobile and can greatly affect the task offloading performance.

To manage the instability of transmissions due to vehicular mobility, numerous studies have focused on delineating the patterns and characteristics of vehicle movements in UVECs. Li et al. [24] analyzed the microscopic motion behavior of vehicles and fully considered the dynamic effect of the vehicle's acceleration and steering angle. Wu et al. [25] assumed that vehicles enter the RSU's coverage area in the scenario based on a Poisson distribution, and the speed of each vehicle is generated by a truncated Gaussian distribution that is flexible and conforms to the real dynamic vehicle environment. Li et al. [26] consider a dynamic MEC network and propose a heuristic-assisted multi-agent reinforcement learning (RL)-based framework to realize the joint optimization of computation offloading and resource allocation. Younis et al. [27] use approximate computing techniques to maximize the available computing resources on the MEC server. However, the above studies only considered the effect of vehicle mobility on task transmission and did not consider the service reliability after the task enters the node queue.

In the analysis of service reliability for delay-sensitive services, the intricate queuing behaviors in UVEC have been explored instead of average metrics, which poses significant difficulties in formulating determinative latency. To better characterize queuing behaviors, SNC theory is widely used as a mathematical tool to analyze queue status [28]. SNC can compute non-asymptotic statistical performance bounds while considering complex stochastic processes. Xiong et al. [19] design an intelligent task offloading framework for millimeter wave (mmWave) communication-based heterogeneous vehicular networks and derive the latency upper bounds of different offloading methods with specific failure probabilities using SNC. Zhu et al. [29] introduced the SNC-based min-plus convolution and the leftover service to analyze the performance and optimization in the Internet-of-Remote Things (IoRT) network. Chen et al. [30] used SNC to model the transmission latency upper bound for platoon-based autonomous driving through cellular V2V communications.

Although various studies have proposed different solutions to meet the demands of services, little literature comprehensively explores how to meet the requirements of latency-sensitive applications in a resource-constrained system with multiple vehicles moving at high speed. Unlike previous work, we study the movement characteristics of vehicles and investigate the stochastic process in UAV systems by the SNC theory. Moreover, few studies have adjusted task throughput to meet the QoS requirements of the system. To guarantee the service reliability for UAVs in the UVEC framework, the maximum throughput with buffer constraints and task delay violation probability are both utilized in our reliability model. Based on the reliability analysis, we propose a distributed algorithm to maximize energy efficiency for latency-sensitive applications in UVEC.

III. SYSTEM MODEL

We consider a UVEC as shown in Fig. 1, which consists of multiple UAVs and vehicles as well as a base station (BS) b

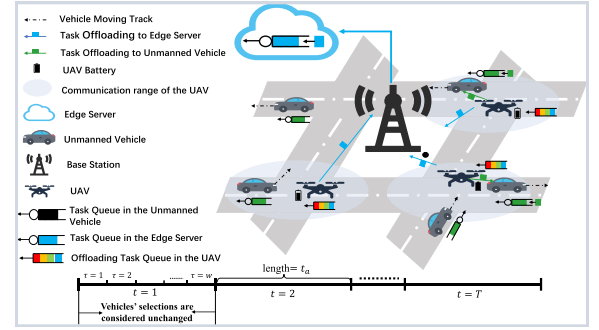


Fig. 1. System overview: a UAV-based vehicular edge computing.

connected to an edge server. The computational capacities of the server or vehicles are f_b and f_v (cycles/s) respectively.

All UAVs in our system model are indexed by $\mathcal{U} = \{i | i = 1, 2, \dots, u\}$ and the vehicle set is treated as $\mathcal{V} = \{j | j = 1, 2, \dots, v\}$. In our model, the system running time can be divided into T time slots, with each time slot having a length of t_a , denoted as $\mathcal{T} = \{t | t = 1, 2, \dots, T\}$. Each time slot t can be divided into a total number of w tiny time intervals τ , and each of the tiny time intervals has the same length of $\frac{t_a}{w}$ as shown in Fig. 1. Each UAV can simultaneously offload tasks to nearby vehicles within its communication coverage or to edge servers through the BS b . UAV selects vehicles in each time slot t . $x_{i,j}(t) \in \{0, 1\}$ is an indicator to denote whether the UAV i selects the vehicle j . A UAV can only connect to one vehicle within t , while a vehicle j can connect to multiple UAVs but with a limited number $A_{\max}$ of connections, just shown below.

$$\begin{cases} \sum_{i \in \mathcal{U}} x_{i,j}(t) \leq A_{\max}, \\ \sum_{j \in \mathcal{V}} x_{i,j}(t) = 1. \end{cases} \quad (1)$$

Each UAV should decide the task offloading ratio between the selected vehicle and the edge server in each large slot. The offloading ratios for transmissions to the vehicle and to the BS b of the UAV i at the time t are denoted as $\xi_i(t)$ and $1 - \xi_i(t)$ respectively. Denote the task throughput from the UAV i to the BS b and from the UAV i to the vehicle j at the time t as $r_{i,b}(t)$ and $r_{i,j}(t)$. The total amount of tasks generated by a UAV during t is $\mathcal{R}$, so $r_{i,b}(t)$ and $r_{i,j}(t)$ are:

$$\begin{cases} r_{i,j}(t) = \xi_i(t)\mathcal{R}, \\ r_{i,b}(t) = (1 - \xi_i(t))\mathcal{R}. \end{cases} \quad (2)$$

In our system model, the BS b is located at the central position. UAVs' locations are modeled as a homogeneous Poisson Point Process (PPP) [31] inside the disc, which means that UAVs' locations follow a uniform probability distribution. The channel gain from the UAV i to the BS b is represented by $h_{i,b}$. Similarly, the channel gain between the UAV i and the vehicle j is denoted by $h_{i,j}$. As illustrated in [32], the ground-to-UAV channel follows the free-space path loss model, so $h_{i,b}$ and $h_{i,j}$ are both determined by the distance. Necessary notations in our formulation are listed in Table I.

TABLE I
 SUMMARY OF MAIN NOTATIONS

Notation	Description	Notation	Description
$r_{i,b}(t)$	Task sending rate between the UAV i and the BS b	$r_{i,j}(t)$	Task sending rate between the UAV i and the vehicle j
$h_{i,b}(t)$	Channel gain between the UAV i and the BS b	$h_{i,j}(t)$	Channel gain between the UAV i and the vehicle j
$\gamma_{b,i}(t)$	SINR between the UAV i and the BS b	$\gamma_{i,j}(t)$	SINR between the UAV i and the vehicle j
$x_{i,j}(t)$	Indicator of whether the UAV i offloads to the vehicle j	$\xi_i(t)$	Task offloading ratio
$e_{i,b}^D(t)$	Delay violation probability for tasks transmitted to the BS b	$e_{i,j}^D(t)$	Delay violation probability for tasks transmitted to the vehicle j
$\epsilon_{i,b}^{buf}$	Buffer overflow probability of tasks transmitted to the server	$\epsilon_{i,j}^{buf}$	Buffer overflow probability of tasks transmitted to the vehicle j
L_b	Buffer constraint for tasks transmitted to the edge server	L_u	Buffer constraint for tasks transmitted to the vehicle j
$\mathcal{E}_{i,b}(t)$	Total energy consumption between the BS b and the UAV i	$\mathcal{E}_{i,j}(t)$	Total energy consumption between the UAV i and the vehicle j
$\mathcal{C}_i$	Communication coverage area of the UAV i	$\Gamma_i(t)$	Service reliability of the UAV i
T	Total time	τ	Tiny time interval
$q_i^u(t)$	Remaining task queue in the UAV i	$q_b(t)$	Task queue in the edge server
$q_j^v(t)$	Task queue in the vehicle j	$\mathcal{R}$	Total amount of tasks generated by each UAV in every time slot
f_b	Computational capacity for the server	f_v	Computational capacity for the vehicle
ι_b	Effective switched capacitance in the edge server depending on the chip architecture	ι_v	Effective switched capacitance in the vehicle depending on the chip architecture
P	Vehicle area transfer matrix	π_i	Steady-state location distribution for vehicles
Ψ_i	Transition matrix of the vehicle's position relative to the communication range of the UAV i	π_i^l	Steady-state distribution of the vehicle's position relative to the communication range of the UAV i

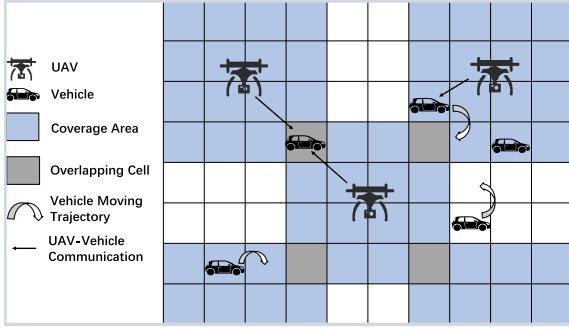


Fig. 2. Illustration of vehicle mobility in a cell-partitioned scenario with overlaps of communication coverage range for UAVs.

A. Mobility Model of Vehicles

As illustrated in Fig. 2, each UAV can only communicate with vehicles within its communication range, which encompasses the same number of cells for each UAV. The coverage areas of different UAVs may overlap, and a vehicle within an overlapping area may be able to connect to multiple UAVs simultaneously. Denotes the communication coverage area of the UAV i as $\mathcal{C}_i = \{c_i^1, \dots, c_i^k\}$, where $\mathcal{C}_i \subseteq \mathcal{C}$. Due to the small distance variation of vehicles within the cell range in a cellular network, the change of channel gain in the same cell between a UAV and a vehicle is tiny and can be ignored [33]. In the initial stage of the UVEC system, vehicles are dispersed randomly across various cells. At the initial stage, a vehicle may be within the communication range of multiple UAVs or outside the range of any UAV. The position of the vehicle may be different between different tiny time intervals. The movement of the vehicle between cells follows a Markov process [34] and is controlled by a Markov

transfer matrix P , which is denoted as follows:

$$P = \begin{bmatrix} p_{1 \rightarrow 1} & p_{1 \rightarrow 2} & \cdots & p_{1 \rightarrow C} \\ p_{2 \rightarrow 1} & p_{2 \rightarrow 2} & \cdots & p_{2 \rightarrow C} \\ \vdots & \vdots & \ddots & \vdots \\ p_{C \rightarrow 1} & p_{C \rightarrow 2} & \cdots & p_{C \rightarrow C} \end{bmatrix}, \quad (3)$$

where $p_{m \rightarrow n}$ is the conditional probability of a vehicle moving to the cell n from the cell m in the current time interval τ .

The vehicle transitions constitute a stable Markov chain, implying that each vehicle's transition dynamics are governed by a fixed, steady-state transition matrix during Markov state transitions within each cell. Denote the steady-state location distribution for every vehicle as $\pi = \{\pi_c\}_{1 \times C}$ for vehicles over the cell $c \in \mathcal{C}$ satisfies $\pi P = \pi$. Thus, the steady-state of vehicle's location distribution π_{C_i} over the communication coverage area $\mathcal{C}_i$ of the UAV i is obtained by:

$$\pi_{C_i} = \sum_{c_i^m \in \mathcal{C}_i} \pi_{c_i^m}. \quad (4)$$

B. Task Transmission Model

As shown in Fig. 3, each UAV uses two queues $q_{i,b}$ and $q_{i,j}$ to store tasks that have not been sent to the server and the vehicle, which are parallel. For the queue in each UAV or vehicle, the arrival process A is the size of tasks arriving at the queue during the current time slot. The service process S is the total data processed by the server in the current time slot, and the departure process A^* is the data that leaves the task queue after being processed. Use $A_{i,b}$ to represent the traffic arrival process of the queue of tasks transmitted to the BS in the UAV i . $A_{i,b}^*$ is the departure process of $q_{i,b}$. The service process for $q_{i,b}$ is denoted as $S_{i,b}$. Denote $A_{i,j}$ as the traffic arrival process between UAV

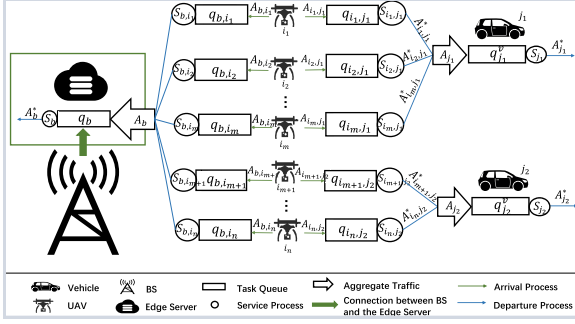


Fig. 3. The series-parallel queuing model for UVEC.

i and vehicle j , and the departure process of $A_{i,j}$ as $A_{i,j}^*$. $S_{i,j}$ is the service process of $q_{i,j}$.

The task queue of the edge server is an aggregate queue of all UAVs and is represented as q_b . Denote A_b as the aggregated arrival process and S_b as the service process of the edge server. A_b^* is used to denote the corresponding departure process of A_b . Denote the task queue in the vehicle j as $q_j^v(t)$. Tasks from UAVs form aggregate traffic $A_j = \sum_{i \in \mathcal{U}} A_{i,j}$ in the vehicle and enter the task queue $q_j^v(t)$.

1) *Task Transmission Model for UAVs and BS*: The UAV $i \in \mathcal{U}$ hovers in a fixed position in the sky and maintains a stable distance from the base station b . Thus, the channel gain $h_{i,b}(t)$ and signal-to-noise ratio (SINR) $\gamma_{i,b}(t)$ in every time slot remain constant. The value of $\gamma_{i,b}$ is obtained as below:

$$\gamma_{i,b} = \frac{\delta_{i,b} |h_{i,b}|^2}{\sigma^2}, \quad (5)$$

where $\delta_{i,b}$ is the transmission power between BS b and UAV i . σ^2 is the variance of Gaussian white noise.

Denote $r_{i,b}(t)$ as the task throughput between the edge server and UAV i , and $R_{i,b}^{thr} = W_b \log(1 + \gamma_{i,b})$ as the maximum channel capacity. $R_{i,b}^{thr}$ is calculated using the Shannon formula [35] and represents the upper capacity limit of an ideal channel without transmission error rate. The cumulative amount of $A_{i,b}$ and $S_{i,b}$ during $[t_0, t_x]$ can be obtained from $A_{i,b}(t_0, t_x) = \sum_{t=t_0}^{t_x} r_{i,b}(t)$ and $S_{i,b}(t_0, t_x) = \sum_{t=t_0}^{t_x} R_{i,b}^{thr} = (t_x - t_0)W_b \log(1 + \gamma_{i,b})$, respectively. W_b in the equation is the bandwidth between a UAV and BS b .

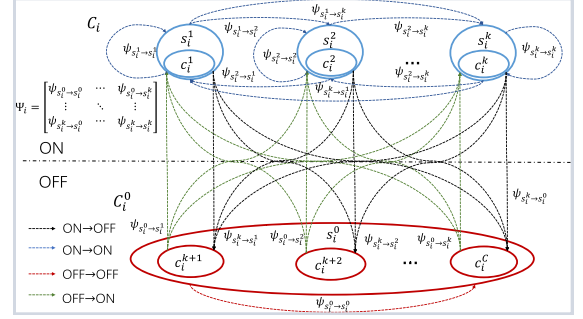
The update of $q_{i,b}$ is obtained by:

$$\begin{aligned} q_{i,b}(t) &= A_{i,b}(0, t) - A_{i,b}^*(0, t) \\ &= \sup_{0 \leq t_0 \leq t} \{A_{i,b}(t_0, t) - S_{i,b}(t_0, t)\}, \end{aligned} \quad (6)$$

where the expression of $A_{i,b}^*$ can be obtained from:

$$A_{i,b}^*(t_0, t) = \inf_{t_0 \leq t_x \leq t} \{A_{i,b}(t_0, t_x) + S_{i,b}(t_x, t)\}. \quad (7)$$

A tuple $(L_b, \epsilon_{i,b}^{buf})$ is used to represent the queue backlog constraint of $q_{i,b}$ as in [28]. L_b is the finite cache capacity for storing backlogged tasks, and $\epsilon_{i,b}^{buf}$ is the buffer overflow probability constraint. The maximal size of the transmission task $R_{i,b}^{max}$ is the maximum constraint that the edge server can support

Fig. 4. The state transition matrix Ψ_i for the relative position between a UAV and a vehicle.

while satisfying L_b . $R_{i,b}^{max}$ can be obtained by:

$$R_{i,b}^{max} = \max\{r_{i,b}(t) : Pr\{q_{i,b}(t) > L_b\} \leq \epsilon_{i,b}^{buf}\}. \quad (8)$$

Combining with (8), the size of the transmission task from the UAV i to the BS is subject to the following constraint:

$$0 \leq r_{i,b}(t) \leq R_{i,b}^{max}, \forall i \in \mathcal{U}. \quad (9)$$

The cumulative arrival process $A_b(t_0, t_x)$ of $q_b(t)$ can be obtained from $A_b(t_0, t_x) = \sum_{i \in \mathcal{U}} A_{i,b}(t_0, t_x)$. The task size serviced in the service process of queue q_b is the accumulation of the processor capacity over the service time. Given that the length of each time slot in the scenario is t_a , the cumulative service process of queue q_b is calculated as $S_b(t_0, t_x) = \sum_{t=t_0}^{t_x} t_a f_b t$. The update of $q_b(t)$ is denoted as below [13]:

$$\begin{aligned} q_b(t) &= A_b(0, t) - A_b^*(0, t) \\ &= \sup_{0 \leq t_0 \leq t} \{A_b(t_0, t) - S_b(t_0, t)\}, \end{aligned} \quad (10)$$

where A_b^* can be obtained from [13] as below:

$$A_b^*(t_0, t) = \inf_{t_0 \leq t_x \leq t} \{A_b(t_0, t_x) + S_b(t_x, t)\}. \quad (11)$$

2) *Task Transmission Model for UAVs and Vehicles*: Denote $\gamma_{i,j}$ as the SINR of the task transmission from the UAV i to the vehicle j which has the expression:

$$\gamma_{i,j} = \frac{\delta_{i,j} |h_{i,j}|^2}{\sigma^2}, \quad (12)$$

where $\delta_{i,j}$ is the power for the transmission between the UAV i and the vehicle j . The channel from the UAV i to the vehicle j is a path-loss model, which is only related to the UAV i and the vehicle j 's communication distance.

When the vehicle's movement follows the Markov process described in the previous section, Ψ_i is used to denote the probability transfer matrix of different communication states from the UAV i to the vehicle that follows this movement model. Additionally, the vehicle's state transition matrix in (3) can be utilized to derive Ψ_i , with the solution process illustrated in Fig. 4. Each element of Ψ_i represents the state transition probability of the channel gain between the UAV and the vehicle.

The transition matrix Ψ_i can be obtained by:

$$\Psi_i = \begin{bmatrix} \psi_{s_i^0 \rightarrow s_i^0} & \psi_{s_i^0 \rightarrow s_i^1} & \cdots & \psi_{s_i^0 \rightarrow s_i^k} \\ \psi_{s_i^1 \rightarrow s_i^0} & \psi_{s_i^1 \rightarrow s_i^1} & \cdots & \psi_{s_i^1 \rightarrow s_i^k} \\ \vdots & \vdots & \ddots & \vdots \\ \psi_{s_i^k \rightarrow s_i^0} & \psi_{s_i^k \rightarrow s_i^1} & \cdots & \psi_{s_i^k \rightarrow s_i^k} \end{bmatrix}, \quad (13)$$

where s_i^m is the communication channel state between UAV i and the vehicle at the cell m , we have $s_i^m \in s_i = \{s_i^0, s_i^1, \dots, s_i^k\}$. Vehicles around the UAV undergo Markov state changes over time. Define $\mathcal{C}_i^0$ as the set of all cells outside the communication range of UAV i . The state s_i^0 indicates that the vehicle is located in a cell within $\mathcal{C}_i^0$, and the communication state with UAV i is "OFF". Conversely, s_i^m for $m \geq 1$ signifies that the vehicle is in cell m and within the communication range of UAV i , thus the communication state is "ON". From Fig. 4 we can see that cells within the communication range of the UAV i are represented by blue circles, while cells outside the communication range are represented by red circles. Correspondingly, when the vehicle moves within the communication range of the UAV, the state transition curve is represented by a blue line. When the vehicle moves outside the communication range of the UAV, the state transition curve is represented by a red line. If the vehicle leaves the communication range of the UAV, the communication state between the vehicle and the UAV changes from "ON" to "OFF", which is indicated by a black line in the figure. For the vehicle enters the communication range of the UAV from outside, the communication state between the vehicle and the UAV changes from "OFF" to "ON", whose transition is represented by a green line.

Denote c_n and $c_{n'}$ as the vehicle's previous located cell and the cell it will transfer next, and denote $\psi_{s_i^n \rightarrow s_i^{n'}}$ as the transition probability of the channel gain in cell c_n and $c_{n'}$. $\psi_{s_i^n \rightarrow s_i^{n'}}$ can be derived from four different cases:

Case 1: If the vehicle's state transition occurs between two cells both within the communication range of the UAV i , i.e. $c_n, c_{n'} \in \mathcal{C}_i$:

$$\psi_{s_i^n \rightarrow s_i^{n'}} = p_{c_n \rightarrow c_{n'}}. \quad (14)$$

Case 2: If the vehicle is within the communication range of the UAV i in the current time slot but moves out of the range in the next time slot, i.e. $c_n \in \mathcal{C}_i, c_{n'} \in \mathcal{C}_i^0$, $\psi_{s_i^n \rightarrow s_i^{n'}}$ can be obtained from:

$$\psi_{s_i^n \rightarrow s_i^{n'}} = \sum_{c_{n'} \in \mathcal{C}_i^0} p_{c_n \rightarrow c_{n'}}. \quad (15)$$

Case 3: If the vehicle is not in the coverage range of the UAV i in the previous time slot and enters it in the next time slot, i.e. $c_n \in \mathcal{C}_i^0, c_{n'} \in \mathcal{C}_i$, $\psi_{s_i^n \rightarrow s_i^{n'}}$ is shown as below:

$$\psi_{s_i^n \rightarrow s_i^{n'}} = \sum_{c_n \in \mathcal{C}_i^0} p_{c_n \rightarrow c_{n'}}. \quad (16)$$

Case 4: If the vehicle moves outside the coverage range of the UAV i in two adjacent time slots, i.e. $c_n, c_{n'} \in \mathcal{C}_i^0$. $\psi_{s_i^n \rightarrow s_i^{n'}}$

can be obtained from:

$$\psi_{s_i^n \rightarrow s_i^{n'}} = \sum_{c_n \in \mathcal{C}_i^0} \sum_{c_{n'} \in \mathcal{C}_i^0} p_{c_n \rightarrow c_{n'}}. \quad (17)$$

Denote the steady-state probability matrix that derived from Ψ_i as $\pi'_i = \{\pi'_c\}_{1 \times (k+1)}$. Based on the above calculations:

$$\pi'_i = \Psi_i \pi'_i. \quad (18)$$

The transition matrix of the vehicle within each UAV's communication range can be viewed as a Markov modulated process (MMP) with multiple states according to the proof in Theorem 1.

Theorem 1: The state transition of the vehicle's position relative to the communication range of the UAV i can be viewed as an MMP. For transition matrix Ψ_i , any cell $m, m', n \in \mathcal{C}$, we have:

$$\psi_{s_i^m \rightarrow s_i^{m'}} \psi_{s_i^{m'} \rightarrow s_i^n} = \psi_{s_i^m \rightarrow s_i^n}.$$

Proof: Please get the detailed proof of Theorem 1 through Appendix A, available online. ■

The cumulative amount of $A_{i,j}$ during $[t_0, t_x]$ is calculated as $A_{i,j}(t_0, t_x) = \sum_{t=t_0}^{t_x} x_{i,j}(t) r_{i,j}(t)$. The cumulative service process can be obtained from $S_{i,j}(t_0, t_x) = \sum_{t=t_0}^{t_x} S_{i,j}(t)$. $S_{i,j}$ can be obtained from:

$$S_{i,j}(t-1, t) = \sum_{\tau=1}^w S_{i,j,t}(\tau). \quad (19)$$

Similar as (11), the expression of $A_{i,j}^*$ can be obtained from:

$$A_{i,j}^*(t_0, t) = \inf_{t_0 \leq t_x \leq t} \{A_{i,j}(t_0, t_x) + S_{i,j}(t_x, t)\}. \quad (20)$$

When the vehicle leaves the communication range of the UAV, the tasks generated by the UAV will not be sent to the vehicle due to channel conditions, resulting in a backlog of UAV tasks. $q_{i,j}$ has a finite cache capacity L_u for storing backlogged tasks. The update of $q_{i,j}$ denotes:

$$\begin{aligned} q_{i,j}(t) &= A_{i,j}(0, t) - A_{i,j}^*(0, t) \\ &= \sup_{0 \leq t_0 \leq t} \{A_{i,j}(t_0, t) - S_{i,j}(t_0, t)\}. \end{aligned} \quad (21)$$

The maximal size of transmission task $R_{i,j}^{\max}$ constrained by $(L_u, \epsilon_{i,j}^{buf})$ is:

$$R_{i,j}^{\max} = \max\{r_{i,j}(t) : Pr\{q_{i,j}(t) > L_u\} \leq \epsilon_{i,j}^{buf}\}, \quad (22)$$

where $\epsilon_{i,j}^{buf}$ is the constraint of the probability that the length of the backlogged task queue $q_{i,j}(t)$ exceeds the capacity L_u .

Similar to (9), the task throughput $r_{i,j}(t)$ is subject to the following constraint:

$$0 \leq r_{i,j}(t) \leq R_{i,j}^{\max}, \forall i \in \mathcal{U}, j \in \mathcal{V}. \quad (23)$$

The channel between the UAV i and the vehicle j is in series with the service queue in the vehicle. Similar to the queue q_b , the task size serviced by the vehicle j is the accumulation of the number of tasks processed by the vehicle's processor over the service time. The cumulative service process in the vehicle j can be obtained from $S_j(t_0, t_x) = \sum_{t=t_0}^{t_x} t_a f_v t$. The

update of the task queue $q_j^v(t)$ in the vehicle j is obtained by:

$$\begin{aligned} q_j^v(t) &= A_j(0, t) - A_j^*(0, t) \\ &= \sup_{0 \leq t_0 \leq t} \{A_j(t_0, t) - S_j(t_0, t)\}. \end{aligned} \quad (24)$$

C. Task Processing Delay Model

Denote $T_{i,k}^{total}(t)$ as the total delay of the task sent at time t to computing node k , k represents the vehicle j or the BS b . $T_{i,k}^{total}(t)$ can be calculated from:

$$T_{i,k}^{total}(t) = T_{i,k}^{tran}(t) + T_{i,k}^{que}(t) + T_{i,k}^{pro}(t), \quad (25)$$

where $T_{i,k}^{tran}(t)$ is the transmission delay for tasks sent at the time t , $T_{i,k}^{que}(t)$ is the queuing delay and $T_{i,k}^{pro}(t)$ is the task processing delay.

Tasks are segmented into different sizes in each time slot and transmitted through wireless channels. Since the UAV's position typically remains unchanged, the channel between the UAV and the BS remains stable, resulting in a tiny change in the free-path loss model. For any UAV $i \in \mathcal{U}$, the task transmission delay $T_{i,b}^{tran}(t)$ between UAV i and BS b is positively correlated with the transmission task $r_{i,b}(t)$. Due to the mobility of the vehicle, the transmission delay $T_{i,j}^{tran}(t)$ is not only related to the transmission task $r_{i,j}(t)$ but also to the dynamic change of the channel state between the UAV and the vehicle. More specifically, according to [13], the transmission delay between the UAV i and the computing node k can be obtained as below:

$$T_{i,k}^{tran}(t) = \inf_{0 \leq t_0} \{t_0 : A_{i,k}(0, t) \geq A_{i,k}^*(0, t + t_0)\}. \quad (26)$$

The processing and queuing delay for the task sent at the time t to computation node k can be obtained by:

$$T_{i,k}^{pro}(t) + T_{i,k}^{que}(t) = \inf_{0 \leq t_0} \{t_0 : A_k(0, t) \geq A_k^*(0, t + t_0)\}. \quad (27)$$

To ensure that the task can be completed in time, the threshold of the total delay $T_{i,k}^{total}(m)$ is denoted by T_{thre} . The delay violation probability $\epsilon_{i,k}^D(t)$ of tasks being served by the computing node k from the UAV i is:

$$\epsilon_{i,k}^D(t) = \Pr \{T_{i,k}^{total}(t) > T_{thre}\}. \quad (28)$$

D. Energy Efficiency Model

The energy efficiency model for each UAV can be discussed from four perspectives as below.

1) *Energy Consumption Model for Transmissions Between the Edge Server and UAVs:* Transmission energy consumption $\mathcal{E}_{i,b}^{tran}(t)$ for the UAV i offloading tasks to the BS b is related to the task throughput $r_{i,b}(t)$. Similar to the work in [36], UAV's transmission energy consumption can be viewed as a function of task throughput, which can expressed as:

$$\mathcal{E}_{i,b}^{tran}(t) = \omega_B T_{i,b}^{tran}(t), \quad (29)$$

where ω_B is the unit energy consumption for transmitting tasks to the BS b .

2) *Energy Consumption Model for Transmissions Between Vehicles and UAVs:* Transmission energy consumption $\mathcal{E}_{i,j}^{tran}(t)$

between the UAV i and the vehicle j is:

$$\mathcal{E}_{i,j}^{tran}(t) = \omega_V x_{i,j}(t) T_{i,j}^{trans}(t), \quad (30)$$

where ω_V is the unit energy consumption for transmitting tasks to the vehicle j .

3) *Energy Consumption Model for Task Processing in the Edge Server:* Task Processing energy consumption $\mathcal{E}_{i,b}^{proc}(t)$ in the edge server is related to the task sending rate $r_{i,b}(t)$ and the computational capacities f_b of the server. From [37], we can calculate $\mathcal{E}_{i,b}^{proc}(t)$ through the following equation:

$$\mathcal{E}_{i,b}^{proc}(t) = \iota_b (T_{i,b}^{pro}(t) + T_{i,b}^{que}(t)) f_b^3, \quad (31)$$

where ι_b is the unit energy consumption for tasks processed in the edge server.

4) *Energy Consumption Model for Task Processing in the Vehicle:* Task Processing energy consumption $\mathcal{E}_{i,j}^{proc}(t)$ is obtained by:

$$\mathcal{E}_{i,j}^{proc}(t) = \iota_v x_{i,j}(t) (T_{i,j}^{pro}(t) + T_{i,j}^{que}(t)) f_v^3, \quad (32)$$

where ι_v is the unit energy consumption for processing tasks in the vehicle j .

The total energy consumption of the task transmission can be obtained from the following equations:

$$\mathcal{E}_{i,b}(t) = \mathcal{E}_{i,b}^{tran}(t) + \mathcal{E}_{i,b}^{proc}(t), \quad (33)$$

and

$$\mathcal{E}_{i,j}(t) = \mathcal{E}_{i,j}^{tran}(t) + \mathcal{E}_{i,j}^{proc}(t). \quad (34)$$

Define η_i as the energy efficiency of the UAV i and can be derived as below:

$$\eta_i(t) = \frac{\sum_{j \in \mathcal{V}} \varsigma_v x_{i,j}(t) r_{i,j}(t) + \varsigma_b r_{i,b}(t)}{\sum_{j \in \mathcal{V}} \mathcal{E}_{i,j}(t) + \mathcal{E}_{i,b}(t)}, \quad (35)$$

where ς_b and ς_v are two constants related to bandwidth profits.

E. Service Reliability Model

Service reliability is the ability and probability of the system to complete the specified function within the specified time and under the specified working conditions [38]. In our reliability formulation, the task reliability is determined by the buffer error rate and the task processing delay violation probability. Define Γ_i as the service reliability of the UAV i and $\Gamma_i(t)$ can be obtained by:

$$\begin{aligned} \Gamma_i(t) &= 1 - (1 - \Gamma_{i,b}(t)) \prod_{j \in \mathcal{V}} (1 - \Gamma_{i,j}(t)) \\ &\approx \Gamma_{i,b}(t) + \sum_{j \in \mathcal{V}} \Gamma_{i,j}(t), \end{aligned} \quad (36)$$

where $\Gamma_{i,b}(t)$ and $\Gamma_{i,j}(t)$ are the reliability related to the task offloading between the UAV i and the edge server and between the UAV i and the vehicle j , and can be obtained from:

$$\begin{cases} \Gamma_{i,b}(t) = (1 - \epsilon_{i,b}^{buf})(1 - \epsilon_{i,b}^D(t)), \\ \Gamma_{i,j}(t) = x_{i,j}(t)(1 - \epsilon_{i,j}^{buf})(1 - \epsilon_{i,j}^D(t)). \end{cases} \quad (37)$$

IV. STOCHASTIC DELAY ANALYSIS AND PROBLEM FORMULATION

A. The Buffer-Constrained Maximum Throughput Analysis

The upper limit of the size of transmission task $R_{i,j}^{\max}$ and $R_{i,b}^{\max}$ is related to the queue capacity $q_{i,b}$ and $q_{i,j}$, which is found by (8) and (22). To compute $R_{i,j}^{\max}$ and $R_{i,b}^{\max}$, we use the SNC method [13] to convert the arrival time and service time to the exponential domain.

Define the moment generating function (MGF) [39] as $\mathcal{M}_F(\theta) \triangleq \mathbb{E}[e^{\theta F}]$, where F is a non-negative random variable and $\theta > 0$ is a free parameter. Therefore, the MGF of the task queue backlogs can be derived as $\mathcal{M}_{q_{i,j}(t)}(\theta_1) = \mathbb{E}[(e^{q_{i,j}(t)})^{\theta_1}]$ and $\mathcal{M}_{q_{i,b}(t)}(\theta_2) = \mathbb{E}[(e^{q_{i,b}(t)})^{\theta_2}]$.

The task arrival process from the UAV i can be viewed as a deterministic process with an uncertain arrival rate $r_{i,j}(t)$. The MGF of the task arrival process for $q_{i,j}$ and $q_{i,b}$ are:

$$\mathcal{M}_{A_{i,j}(t-1,t)}(\theta_1) = e^{\theta_1 r_{i,j}(t)}, \quad (38)$$

$$\mathcal{M}_{A_{i,b}(t-1,t)}(\theta_2) = e^{\theta_2 r_{i,b}(t)}. \quad (39)$$

As shown in Theorem 1, the state transition of the vehicle position relative to the UAV communication range i follows a Markov characteristic. The channel between the UAV and the vehicle varies with the movement characteristics of the vehicle. The service process of the channel is a non-independent stochastic process. Similar to [39], the slope $(\rho(\theta), \sigma(\theta))$ -bounded process of the service process of the channel from the UAV i to the vehicle j in each tiny time interval that follows MMP traffic can be obtained by:

$$\begin{cases} \sigma_{S_{i,j}}(\theta_1) = 0, \\ \rho_{S_{i,j}}(\theta_1) = \frac{1}{\theta_1} \log(sp(\Xi(\theta_1)\Psi_i)), \end{cases} \quad (40)$$

where $sp(\cdot)$ is the spectral radius of the matrix, and $\Xi(\theta_1)$ is calculated by:

$$\Xi(\theta_1) = \begin{bmatrix} e^{\theta_1 R_{s_i^0}} & 0 & \cdots & 0 \\ 0 & e^{\theta_1 R_{s_i^1}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{\theta_1 R_{s_i^k}} \end{bmatrix}, \quad (41)$$

$R_{s_i^n}$ in the expression is the maximum throughput when the channel gain state is s_i^n . $R_{s_i^n}$ can be obtained by:

$$R_{s_i^n} = W_v \log(1 + \gamma_{s_i^n}), \quad (42)$$

where W_v in the equation is the bandwidth between a UAV and a vehicle and $\gamma_{s_i^n}$ is the SINR when the channel gain state is s_i^n that can be obtained from (12).

From (19), the total service process in the large time slot can be obtained by:

$$S_{i,j}(t-1, t) = \sum_{\tau=1}^w S_{i,j,t}(\tau) = w\rho_{S_{i,j}}(\theta_1). \quad (43)$$

Based on the above content, the detailed derivation of the maximal size of transmission task $R_{i,j}^{\max}$ and $R_{i,b}^{\max}$ are given in Lemma 1:

Lemma 1: The maximum size of transmission task $R_{i,j}^{\max}$ and $R_{i,b}^{\max}$ can be obtained by:

$$R_{i,j}^{\max} = \frac{L_u \mathbb{E} \left[e^{\frac{\ln \epsilon_{i,j}^{buf}}{L_u} - w\rho_{S_{i,j}}(\theta_1)} \right]}{\ln \epsilon_{i,j}^{buf}}, \quad (44)$$

where $\rho_{S_{i,j}}$ is obtained in (40), and

$$R_{i,b}^{\max} = \frac{L_b \mathbb{E} \left[e^{\frac{\ln \epsilon_{i,b}^{buf}}{L_b} R_{i,b}^{thr}} \right]}{\ln \epsilon_{i,b}^{buf}}. \quad (45)$$

Proof: Please get the detailed proof of Lemma 1 through Appendix B, available online. ■

B. Task Processing Delay Violation Probability Analysis

From (25) and (28), for the computation node k , the upper bound of delay violation probability can be derived as:

$$\epsilon_{i,k}^D(t) \leq e^{-\theta T_{thre}} e^{\theta T_{i,k}^{total}(t)}. \quad (46)$$

The upper bound of latency violation probability of the task transmitted at the time t from the UAV i to the edge server and from the UAV i to the vehicle j can be obtained from Lemma 2:

Lemma 2: The upper bound of latency violation probability of the task transmitted at the time t from the UAV i to the edge server can be derived as:

$$\epsilon_{i,b}^D(t) \leq \frac{e^{\theta_2 \left(\frac{r_{i,b}(t)}{R_{i,b}^{thr}} + \sum_{i \in \mathcal{U}} \frac{[R_{i,b}^{\max} + R_{i,b}^{thr}]}{t_a f_b} - T_{thre} \right)}}{\left[1 - e^{\theta_2 \left(\frac{R_{i,b}^{\max}}{R_{i,b}^{thr}} - 1 \right)} \right] \left[1 - e^{\theta_2 \left(\frac{\sum_{i \in \mathcal{U}} [R_{i,b}^{\max} + R_{i,b}^{thr}]}{t_a f_b} - 1 \right)} \right]},$$

and

$$\epsilon_{i,j}^D(t) \leq \frac{e^{\theta_1 \left(\frac{x_{i,j}(t)r_{i,j}(t)}{w\rho_{S_{i,j}}(\theta_1)} + \sum_{i \in \mathcal{U}} \frac{x_{i,j}(t)[R_{i,j}^{\max} + w\rho_{S_{i,j}}(\theta_1)]}{t_a f_v} - T_{thre} \right)}}{\left[1 - e^{\theta_1 \left(\frac{R_{i,j}^{\max}}{w\rho_{S_{i,j}}(\theta_1)} - 1 \right)} \right] \left[1 - e^{\theta_1 \left(\frac{A_{\max} [R_{i,j}^{\max} + w\rho_{S_{i,j}}(\theta_1)]}{t_a f_v} - 1 \right)} \right]}.$$

Proof: Please get the detailed proof of Lemma 2 through Appendix C, available online. ■

C. Problem Formulation

To maximize energy efficiency while ensuring that all tasks can be transmitted from UAVs within a specified time, service reliability constraints need to be considered. The problem is set below:

$$\mathbf{P1} : \max_{x_{i,j}, \xi_i} \sum_{t=0}^T \sum_{i \in \mathcal{U}} \eta_i(t), \quad (47)$$

s.t. (1), (2), (8), (9), (10), (23), (21), (22), (24),

$$\Gamma_i(t) \geq \Gamma_{\min}, \forall i \in \mathcal{U}, \quad (48)$$

in which (1) is the constraint on connections and $r_i(t)$ is the task size sent by the UAV i at the time t . Equations (9) and (23) is the task transmission rate constraints. Equation (21) is the backlogged task queue in the UAV i due to the channel condition. Equations (10) and (24) are the updates of q_b and q_j^v , respectively. Equations (8) and (22) are the maximum task sending rate constraints. Equation (28) is the delay violation probability constraint. Equation (48) is the reliability constraint for each UAV.

V. DESIGN OF DYNAMIC DISTRIBUTED ALGORITHM

A dynamic distributed algorithm is designed as a solution for **P1** in this section. Since UVEC is a distributed environment, we design a distributed algorithm for energy efficiency optimization. It can be observed that **P1** is not a convex problem. To obtain a near-optimal solution, we employ a linear transformation method to transform the problem. After that, we propose a distributed optimization framework based on the consensus ADMM algorithm to solve **P1**.

A. Problem Transformation

Due to the high computational complexity of **P1**, it is difficult to get the optimal result. As a result, the following transformations are used to reduce the problem's complexity:

1) The Transformation for the Fractional Programming Problem: **P1** is in the form of fractional programming and belongs to a non-convex problem. To achieve the optimal solution of the **P1**, the method of linear transformation that is used in [40] is adopted.

Similar to [40], if $(x_{i,j}^*, \xi_i^*)$ is an optimal solution to **P1**, then there exists an optimal solution $\varepsilon_i = \varepsilon_i^*$ subject to the constraints of (1), (2), (8), (9), (10), (23), (21), (22), (24) and (48). Consequently, $(x_{i,j}^*, \xi_i^*)$ is a solution to the following problem:

$$\begin{aligned} \max_{x_{i,j}, \xi_i} \quad & \sum_{t=0}^T \sum_{i \in \mathcal{U}} \left(\sum_{j \in \mathcal{V}} s_v x_{i,j}(t) r_{i,j}(t) + s_b r_{i,b}(t) \right) \\ & - \varepsilon_i \left(\sum_{j \in \mathcal{V}} \mathcal{E}_{i,j}(t) + \mathcal{E}_{i,b}(t) \right), \end{aligned} \quad (49)$$

and $(x_{i,j}^*, \xi_i^*)$ also satisfies the following system of equations for ε_i^* :

$$\varepsilon_i^* = \frac{\sum_{j \in \mathcal{V}} s_v x_{i,j}^*(t) r_{i,j}^*(t) + s_b r_{i,b}^*(t)}{\sum_{j \in \mathcal{V}} \mathcal{E}_{i,j}^*(t) + \mathcal{E}_{i,b}^*(t)}, \quad \forall i \in \mathcal{U}, \quad (50)$$

The fractional programming objective function $\mathcal{F}^t(\varepsilon_i)$ for (49) in each large time slot can be obtained from:

$$\begin{aligned} \mathcal{F}^t(\varepsilon_i) = \sum_{i \in \mathcal{U}} \left(\sum_{j \in \mathcal{V}} s_v x_{i,j}(t) r_{i,j}(t) + s_b r_{i,b}(t) \right) \\ - \varepsilon_i \left(\sum_{j \in \mathcal{V}} \mathcal{E}_{i,j}(t) + \mathcal{E}_{i,b}(t) \right). \end{aligned} \quad (51)$$

Algorithm 1: The Linear Transformation Algorithm.

Input: $x_{i,j}(t)$, $r_{i,j}(t)$, $\mathcal{E}_{i,j}(t)$, $r_{i,b}(t)$, $\mathcal{E}_{i,b}(t)$: The value of variables at the time t .

$\varepsilon_i(0) = 0$: Initial value of ε_i .

$l = 0, l_{\max}$: Loop step and maximum loop value.

$\varkappa$: Tolerant error.

Output: ε_i^* : Final value of ε_i .

1: $\varepsilon \leftarrow \varepsilon_i(0)$

2: $l \leftarrow 0$

3: **while** $|\mathcal{F}^t(\varepsilon(l)) - \mathcal{F}^t(\varepsilon(l-1))| > \varkappa$ and $l < l_{\max}$ **do**

4: $\varepsilon_{\text{prev}} \leftarrow \varepsilon$

5: Solve equation corresponding to $\varepsilon_{\text{prev}}$ for ε

6: $l \leftarrow l + 1$

7: **end while**

8: $\varepsilon_i^* \leftarrow \varepsilon$

9: **return** ε_i^*

The detail of the linear transformation algorithm for (49) is shown in Algorithm 1.

2) *Binary Variable Relaxation:* $x_{i,j}(t)$ is a binary variable, which makes **P1** hard to handle. As a result, we relax $x_{i,j}(t)$ as a continuous variable, where $0 \leq x_{i,j}(t) \leq 1$.

3) *Global Variable Decomposition and Substitution of P1:* It can be obtained from (1) that $x_{i,j}(t)$ is a global variable, so **P1** is not separable. Denote the local copies of the global variable $x_{i,j}(t)$ for UAV i as $\widetilde{x}_{i,j}(t)$. To address the convexity brought by the product relationships between $x_{i,j}(t)$ and $r_{i,j}(t)$, define $\widetilde{r}_{i,j}(t) = \widetilde{x}_{i,j}(t) r_{i,j}(t) = \widetilde{x}_{i,j}(t) \xi_i(t) \mathcal{R}$.

As a result, combining with (51), **P1** can be changed into the following problem:

$$\begin{aligned} \mathbf{P2} : \quad & \max_{\widetilde{x}_{i,j}, r_{i,b}, \widetilde{r}_{i,j}} \sum_{t=0}^T \sum_{i \in \mathcal{U}} \left(\sum_{j \in \mathcal{V}} s_v \widetilde{x}_{i,j}(t) \widetilde{r}_{i,j}(t) + s_b r_{i,b}(t) \right) \\ & - \varepsilon_i \left(\sum_{j \in \mathcal{V}} \widetilde{\mathcal{E}}_{i,j}(t) + \mathcal{E}_{i,b}(t) \right), \end{aligned} \quad (52)$$

s.t. (1), (8), (9), (10), (21), (24),

$$0 \leq \widetilde{x}_{i,j}(t) \leq 1, \forall i \in \mathcal{U}, \forall j \in \mathcal{V}, \quad (53)$$

$$\widetilde{r}_{i,j}(t) \leq r_{i,j}(t), \quad (54)$$

$$0 \leq \widetilde{r}_{i,j}(t) \leq R_{i,j}^{\max}, \forall i \in \mathcal{U}, j \in \mathcal{V}, \quad (55)$$

$$\widetilde{r}_{i,j}(t) + r_{i,b}(t) = \mathcal{R}, \quad (56)$$

$$\widetilde{\Gamma}_i(t) \geq \Gamma_{\min}, \forall i \in \mathcal{U}, \quad (57)$$

where $\widetilde{\mathcal{E}}_{i,j}(t)$ can be obtained by:

$$\widetilde{\mathcal{E}}_{i,j}(t) = \mathcal{E}_{i,j}(\widetilde{r}_{i,j}(t)).$$

To calculate the total reliability Γ_i for the UAV i , global information about the entire scene is required. However, the computational load and the complexity of the algorithm increase significantly.

To simplify calculations and reduce the computational complexity, we have implemented the following approach:

$$\begin{cases} \sum_{i \in \mathcal{U}} r_{i,b}(t) \approx \overline{ur_{i,b}(t)}, \\ \sum_{i \in \mathcal{U}} x_{i,j}(t) \approx \overline{ux_{i,j}(t)}, \end{cases} \quad (58)$$

where $\overline{r_{i,b}(t)}$ is the average value of the task sending rate between the UAV i and the BS b . $\tilde{\Gamma}_i(t)$ is obtained by:

$$\begin{aligned} \tilde{\Gamma}_i(t) &= (1 - \epsilon_{i,b}^{buf})(1 - \overline{\epsilon_{i,b}^D}(t)) \\ &+ \sum_{j \in \mathcal{V}} \widetilde{x_{i,j}(t)}(1 - \epsilon_{i,j}^{buf})(1 - \widetilde{\epsilon_{i,j}^D}(t)), \end{aligned} \quad (59)$$

where

$$\overline{\epsilon_{i,b}^D}(t) = \frac{e^{\theta_2 \frac{\overline{ur_{i,b}(t)}}{f_b t_a} - \theta_2 T_{thre}}}{1 - e^{\theta_2 \left(\frac{UR_{i,b}^{\max}}{f_b t_a} - 1 \right)}}, \quad (60)$$

and

$$\begin{aligned} \widetilde{\epsilon_{i,j}^D}(t) &= \frac{\theta_1 \left(\frac{\widetilde{r_{i,j}(t)}}{\rho_{S_{i,j}}(\theta_1)} + \frac{ux_{i,j}(t) \left[\frac{R_{i,j}^{\max} + w\rho_{S_{i,j}}(\theta_1)}{t_a f_v} \right] - T_{thre}}{1 - e^{\theta_1 \left(\frac{R_{i,j}^{\max}}{\rho_{S_{i,j}}(\theta_u)} - 1 \right)}} \right)}{\left[1 - e^{\theta_1 \left(\frac{R_{i,j}^{\max}}{\rho_{S_{i,j}}(\theta_u)} - 1 \right)} \right] \left[1 - e^{\theta_1 \left(\frac{A_{\max} \left[\frac{R_{i,j}^{\max} + w\rho_{S_{i,j}}(\theta_1)}{t_a f_v} \right] - 1 \right)} \right]}. \end{aligned} \quad (61)$$

The aforementioned transformation converts **P2** into a convex optimization problem, yet the computational complexity remains high. To streamline the solution of the distributed optimization problem, this paper introduces the Consensus ADMM Algorithm [41].

B. Distributed Optimization Via Consensus ADMM Algorithm

According to [41], the augmented Lagrangian function of **P2** is given as:

$$\begin{aligned} \mathcal{L}(\widetilde{x_{i,j}(t)}, x_{i,j}(t), r_{i,b}(t), r_{i,j}(t), \widetilde{r_{i,j}(t)}, \lambda_i, \mu_i, \nu_i) &= \sum_{i \in \mathcal{U}} \left(\sum_{j \in \mathcal{V}} \varsigma_v x_{i,j}(t) r_{i,j}(t) + \varsigma_b r_{i,b}(t) - \varepsilon_i \left(\sum_{j \in \mathcal{V}} \mathcal{E}_{i,j}(t) \right. \right. \\ &+ \left. \left. \mathcal{E}_{i,b}(t) \right) \right) + \sum_{i \in \mathcal{U}} \lambda_i (\widetilde{x_{i,j}(t)} - x_{i,j}(t)) + \sum_{i \in \mathcal{U}} \mu_i (\widetilde{r_{i,j}(t)} \\ &- r_{i,j}(t)) + \sum_{i \in \mathcal{U}} \nu_i (\Gamma_{\min} - \tilde{\Gamma}_i(t)) + \sum_{i \in \mathcal{U}} \frac{\kappa}{2} (\widetilde{x_{i,j}(t)} - x_{i,j}(t))^2 \\ &+ \sum_{i \in \mathcal{U}} \frac{\kappa}{2} (\widetilde{r_{i,j}(t)} - r_{i,j}(t))^2 + \sum_{i \in \mathcal{U}} \frac{\kappa}{2} (\Gamma_{\min} - \tilde{\Gamma}_i(t))^2, \end{aligned} \quad (62)$$

where λ_i , μ_i and ν_i are the Lagrange multipliers, and κ is the penalty parameter.

1) *Global Variables Update:* Denote $X_i = \{r_{i,b}, \widetilde{x_{i,j}}, \widetilde{r_{i,j}}\}$, $Y_i = \{x_{i,j}, r_{i,j}, \widetilde{r_{i,b}}\}$ and $Z_i = \{\lambda_i, \mu_i, \nu_i\}$, $\mathcal{F}_i(X_i)$ can be calculated as:

$$\begin{aligned} \mathcal{F}_i(X_i) &= \sum_{j \in \mathcal{V}} \varsigma_v x_{i,j}(t) r_{i,j}(t) + \varsigma_b r_{i,b}(t) \\ &- \varepsilon_i \left(\sum_{j \in \mathcal{V}} \mathcal{E}_{i,j}(t) + \mathcal{E}_{i,b}(t) \right). \end{aligned} \quad (63)$$

The updating process of dual variables in $g + 1$ iterations is shown as follows.

$$\begin{aligned} \{X_i\}^{(g+1)} &= \arg \max \left\{ \mathcal{F}_i(X_i) + \lambda_i^{(g)} (\widetilde{x_{i,j}(t)} - x_{i,j}(t))^{(g)} \right. \\ &+ \mu_i^{(g)} (\widetilde{r_{i,j}(t)} - r_{i,j}(t))^{(g)} + \nu_i^{(g)} \sum_{j \in \mathcal{V}} \widetilde{x_{i,j}(t)}(1 - \epsilon_{i,j}^{buf}) \\ &(1 - \widetilde{\epsilon_{i,j}^D}(t)) + \frac{\kappa}{2} [\widetilde{x_{i,j}(t)} - x_{i,j}(t)]^{(g)2} + \frac{\kappa}{2} [\widetilde{r_{i,j}(t)} \\ &- r_{i,j}(t)]^{(g)2} + \frac{\kappa}{2} [(\Gamma_{\min} - \tilde{\Gamma}_i(t))^{(g+1)}]^2 \left. \right\}, \end{aligned} \quad (64)$$

$$\begin{aligned} \{Y_i\}^{(g+1)} &= \arg \max \left\{ \sum_{i \in \mathcal{U}} \lambda_i^{(g)} (\widetilde{x_{i,j}(t)}^{(g+1)} - x_{i,j}(t)) \right. \\ &+ \sum_{i \in \mathcal{U}} \mu_i^{(g)} (\widetilde{r_{i,j}(t)}^{(g+1)} - r_{i,j}(t)) + \sum_{i \in \mathcal{U}} \nu_i^{(g)} (\Gamma_{\min} \\ &- \tilde{\Gamma}_i(t))^{(g+1)} + \frac{\kappa}{2} \sum_{i \in \mathcal{U}} [\widetilde{x_{i,j}(t)}^{(g+1)} - x_{i,j}(t)]^2 \\ &+ \frac{\kappa}{2} \sum_{i \in \mathcal{U}} [(\widetilde{r_{i,j}(t)}^{(g+1)} - r_{i,j}(t))]^2 + \frac{\kappa}{2} \sum_{i \in \mathcal{U}} [(\Gamma_{\min} \\ &- \tilde{\Gamma}_i(t))^{(g+1)}]^2 \left. \right\}, \end{aligned} \quad (65)$$

and the update of $\{Z_i^{(m+1)}\}$ can be obtained by:

$$\begin{cases} \lambda_i^{(g+1)} = \lambda_i^{(g)} + \kappa (\widetilde{x_{i,j}(t)} - x_{i,j}(t))^{(g+1)}, \\ \mu_i^{(g+1)} = \mu_i^{(g)} + \kappa (\widetilde{r_{i,j}(t)} - r_{i,j}(t))^{(g+1)}, \\ \nu_i^{(g+1)} = \nu_i^{(g)} + \kappa (\Gamma_{\min} - \tilde{\Gamma}_i(t))^{(g+1)}. \end{cases} \quad (66)$$

2) *Local Variables Update:* After the global variables are updated, each UAV needs to update the local variables. From (54) and (56), $\widetilde{r_{i,j}(t)}$ can be replaced by $\mathcal{R} - r_{i,b}(t)$. Each UAV needs to solve the following optimization **P2.1** at iteration $m + 1$:

$$\begin{aligned} \mathbf{P2.1} : \max_{\widetilde{x_{i,j}}, r_{i,b}} &\left\{ \mathcal{F}_i(X_i) + \lambda_i^{(g)} (\widetilde{x_{i,j}(t)} - x_{i,j}(t))^{(g)} \right. \\ &+ \mu_i^{(g)} (\mathcal{R} - r_{i,b}(t) - r_{i,j}(t))^{(g)} + \nu_i^{(g)} \sum_{j \in \mathcal{V}} \widetilde{x_{i,j}(t)} \\ &(1 - \epsilon_{i,j}^{buf})(1 - \widetilde{\epsilon_{i,j}^D}(t)) + \frac{\kappa}{2} [\widetilde{x_{i,j}(t)} - x_{i,j}(t)]^{(g)2} + \frac{\kappa}{2} [\mathcal{R} \\ &- r_{i,b}(t) - r_{i,j}(t)]^{(g)2} + \frac{\kappa}{2} [(\Gamma_{\min} - \tilde{\Gamma}_i(t))^{(g+1)}]^2 \left. \right\}, \\ \text{s.t. } &(1), (8), (9), (10), (21), (24), (53), (54), (55), (56), (57). \end{aligned} \quad (67)$$

Note that **P2.1** is a convex problem, so convex optimization methods can obtain the corresponding optimal solution.

To simplify the calculations, we decompose the (67) into different parts and replace the original terms with different letters. According to (64)–(66), we can get that:

$$\begin{cases} \mathcal{G}_i(\widetilde{x}_{i,j}(t)) = \lambda_i^{(g)} (\widetilde{x}_{i,j}(t) - x_{i,j}(t)^{(g)}) + \frac{\kappa}{2} [\widetilde{x}_{i,j}(t) - x_{i,j}(t)^{(g)}]^2, \\ \mathcal{H}_i(r_{i,b}(t)) = \frac{\kappa}{2} [\mathcal{R} - r_{i,b}(t) - r_{i,j}(t)^{(g)}]^2, \\ \mathcal{I}_i(X_i) = \mathcal{F}_i(X_i) + \nu_i^{(g)} \sum_{j \in \mathcal{V}} \widetilde{x}_{i,j}(t) (1 - \epsilon_{i,j}^{buf}) (1 - \epsilon_{i,j}^{\overline{D}}(t)) \\ \quad + \frac{\kappa}{2} [(\Gamma_{\min} - \widetilde{\Gamma}_i(t)^{(g+1)})]^2. \end{cases} \quad (68)$$

So **P2.1** can be transformed into:

$$\mathbf{P2.2} : \max_{\widetilde{x}_{i,j}, r_{i,b}} \{ \mathcal{G}(\widetilde{x}_{i,j}(t)) + \mathcal{H}(r_{i,b}(t)) + \mathcal{I}_i(X_i) \}. \quad (69)$$

3) *Global Variables and Lagrange Multipliers Update*: It can be easily obtained that (65) is an unconstrained quadratic problem and strictly convex. Given $X_i^{(m+1)}$, the global variables $x_{i,j}(t)$, $r_{i,j}(t)$ and $\overline{r_{i,b}(t)}$ can be obtained as follows, respectively:

$$\begin{aligned} \left\{ \sum_{i \in \mathcal{U}} x_{i,j}(t) \right\}^{(g+1)} &= \arg \max \left\{ \frac{\kappa}{2} \sum_{i \in \mathcal{U}} [\widetilde{x}_{i,j}(t)^{(g+1)} - x_{i,j}(t)]^2 \right. \\ &\quad \left. + \sum_{i \in \mathcal{U}} \lambda_i^{(g)} (\widetilde{x}_{i,j}(t)^{(g+1)} - x_{i,j}(t)) \right\}, \end{aligned} \quad (70)$$

$$\begin{aligned} \left\{ \sum_{i \in \mathcal{U}} r_{i,j}(t) \right\}^{(g+1)} &= \arg \max \left\{ \frac{\kappa}{2} \sum_{i \in \mathcal{U}} [\widetilde{r_{i,j}(t)}^{(g+1)} - r_{i,j}(t)]^2 \right. \\ &\quad \left. + \sum_{i \in \mathcal{U}} \mu_i^{(g)} (\widetilde{r_{i,j}(t)}^{(g+1)} - r_{i,j}(t)) \right\}, \end{aligned} \quad (71)$$

$$\begin{aligned} \left\{ \overline{r_{i,b}(t)} \right\}^{(g+1)} &= \arg \max \left\{ \frac{\kappa}{2} \sum_{i \in \mathcal{U}} [(\Gamma_{\min} - \widetilde{\Gamma}_i(t)^{(g+1)})]^2 \right. \\ &\quad \left. + \sum_{i \in \mathcal{U}} \nu_i^{(g)} (\Gamma_{\min} - \widetilde{\Gamma}_i(t)^{(g+1)}) \right\}. \end{aligned} \quad (72)$$

$\widetilde{\Gamma}_i(t)^{(g+1)}$ in (65) is calculated by:

$$\begin{aligned} \widetilde{\Gamma}_i(t)^{(g+1)} &= (1 - \epsilon_{i,b}^{buf}) (1 - \overline{\epsilon_{i,b}^D}(t)^{(g)}) \\ &\quad + \sum_{j \in \mathcal{V}} \widetilde{x_{i,j}(t)}^{(g+1)} (1 - \epsilon_{i,j}^{buf}) (1 - \widetilde{\epsilon_{i,j}^D}(t)^{(g+1)}). \end{aligned}$$

After obtaining $X_i^{(g+1)}$ and $Y_i^{(g+1)}$ from the above equations, Lagrange multipliers λ_i , μ_i and ν_i in the $(g+1)$ th iteration are derived from (66).

To sum up, the details of the proposed consensus-ADMM based algorithm are shown in Algorithm 2.

Algorithm 2: The Consensus-ADMM Based Algorithm.

Input: $r_{i,b}(t)^{(0)}$, $x_{i,j}(t)^{(0)}$, and $r_{i,j}(t)^{(0)}$: Initial values at time t .
 $g = 0$, $g_{\max}$: Initial loop step and maximum loop value.
 $\varphi_x, \varphi_y, \varphi_z$: Stopping criterion thresholds for λ_i, μ_i , and ν_i .
 $\lambda_i^{(0)}, \mu_i^{(0)}, \nu_i^{(0)}$: Initial values of Lagrange multipliers.
Output: $r_{i,b}^*(t)$, $x_{i,j}^*(t)$, and $r_{i,j}^*(t)$: Optimal values at time t .
while $\| \widetilde{x_{i,j}(t)} - x_{i,j}(t) \|_2^{(g)} > \varphi_x$ or $\| \widetilde{r_{i,j}(t)} - r_{i,j}(t) \|_2^{(g)} > \varphi_y$ or $\| \Gamma_{\min} - \widetilde{\Gamma}_i(t) \|_2^{(g)} > \varphi_z$ and $g < g_{\max}$ **do**
 2: Solve **P2.2** to obtain local optimal variables $\widetilde{x_{i,j}(t)}^{(g)}$, $\widetilde{r_{i,j}(t)}^{(g)}$, and $\widetilde{r_{i,b}(t)}^{(g)}$.
 Obtain global optimal variables using (70)–(72): $\sum_{i \in \mathcal{U}} x_{i,j}(t)^{(g)}$, $\sum_{i \in \mathcal{U}} r_{i,j}(t)^{(g)}$, and $\overline{r_{i,b}(t)}^{(g)}$.
 4: Update Lagrange multipliers $\lambda_i^{(g)}, \mu_i^{(g)}, \nu_i^{(g)}$ using updates specified in Z_i .
 $g \leftarrow g + 1$.
end while
return $r_{i,b}^*(t)$, $x_{i,j}^*(t)$, and $r_{i,j}^*(t)$

C. Joint Task Offloading Decision and Offloading Ratio Algorithm (JTODAOR)

After employing the aforementioned algorithm, we obtain the corresponding offloading ratios of each UAV and transmission task amounts for vehicles and the server. By using the rounding principle mentioned in [42], the binary variable $x_{i,j}(t)$ is obtained.

Our proposed strategy to solve **P1** is summarized in Algorithm 3.

D. Performance Analysis

1) *Algorithm Complexity*: Algorithm 1 is iterated $l_{\max}$ times, with each iteration having a computational complexity of $O(u)$. Thus, the total time complexity of Algorithm 1 is $O(ul_{\max})$. In contrast, the number of iterations for Algorithm 2 is primarily constrained by $g_{\max}$, and each iteration has a complexity of $O(uv)$. Consequently, the total time complexity of Algorithm 2 is $O(uvg_{\max})$. Taking into account the complexities of these two sub-algorithms, along with the additional operations in JTODAOR, the overall complexity of our algorithm is $O(T \cdot (ul_{\max} + uvg_{\max} + v))$.

2) *The Convergence of the Strategy*: To demonstrate the convergence of the strategy, the following theorems are proposed and proven.

Theorem 2: Our proposed strategy can be proved objective function value convergence, and we have:

$$\begin{aligned} \sum_{i \in \mathcal{U}} \mathcal{F}_i(X_i^*) - \sum_{i \in \mathcal{U}} \mathcal{F}_i(X_i^{(g+1)}) &\leq \sum_{i \in \mathcal{U}} \lambda_i^{(*)} (\widetilde{x_{i,j}(t)} - x_{i,j}(t))^{(g+1)} \\ &\quad + \sum_{i \in \mathcal{U}} \mu_i^{(*)} (\widetilde{r_{i,j}(t)} - r_{i,j}(t))^{(g+1)} \end{aligned}$$

Algorithm 3: Joint Task Offloading Decision and Offloading Ratio Algorithm (JTODAOR).

Initialization: Set $t = 0$, initialize $q_b(0) = 0$, and $q_j(0) = 0$ for all $j \in \mathcal{V}$.
while $t \leq T$ **do**
 3: Transform the fractional programming part in **P1** using linear transformation by Algorithm 1. Obtain the offloading ratio and task throughput $r_{i,b}(t)$ and $r_{i,j}(t)$ for the UAV i to the BS b and the vehicle j respectively, using Algorithm 2. Determine the binary variable $x_{i,j}(t)$ using the rounding principle.
 6: Update the queue lengths $q_b(t)$ and $q_j(t)$ for all $j \in \mathcal{V}$. Increment time step: $t = t + 1$.
end while

$$+ \sum_{i \in \mathcal{U}} \nu_i^{(*)} (\Gamma_{\min} - \tilde{\Gamma}_i(t)^{(g+1)}).$$

Proof: Please get the detailed proof of Theorem 2 through Appendix D, available online. ■

The algorithm proposed in Section V can be proved dual variable convergence in Theorem 3.

Theorem 3: Our proposed algorithm can be proved dual variable convergence for all UAVs, we have:

$$\begin{aligned} & \sum_{i \in \mathcal{U}} \mathcal{F}_i(X_i^{(g+1)}) - \sum_{i \in \mathcal{U}} \mathcal{F}_i(X_i^*) \leq \sum_{i \in \mathcal{U}} (\widetilde{x}_{i,j}(t)^* \\ & - \widetilde{x}_{i,j}(t)^{(g+1)}) [\lambda_i^{(g+1)} - \kappa \sum_{i \in \mathcal{U}} (x_{i,j}(t)^{(g+1)} - x_{i,j}(t)^{(g)})] \\ & + \sum_{i \in \mathcal{U}} (\widetilde{r}_{i,j}(t)^* - \widetilde{r}_{i,j}(t)^{(g+1)}) [\mu_i^{(g+1)} - \kappa \sum_{i \in \mathcal{U}} (r_{i,j}(t)^{(g+1)} \\ & - r_{i,j}(t)^{(g)})] + \sum_{i \in \mathcal{U}} (\tilde{\Gamma}_i(t)^* - \tilde{\Gamma}_i(t)^{(g+1)}) [\nu_i^{(g+1)} - \kappa \\ & \cdot \sum_{i \in \mathcal{U}} (\tilde{\Gamma}_i(t)^{(g+1)} - \Gamma_{\min})]. \end{aligned}$$

Proof: Please get the detailed proof of Theorem 3 through Appendix E, available online. ■

Define a Lyapunov function B for **P2**, the value of B for the $g + 1$ iteration is obtained by:

$$B^{(g+1)} = \frac{1}{\kappa} \|Z_i^{(g+1)} - Z_i^*\|_2^2 + \kappa \|Y_i^{(g+1)} - Y_i^*\|_2^2. \quad (73)$$

In Theorem 4, we can prove that our algorithm is residual convergence.

Theorem 4: Based on (73), our algorithm can be proved residual convergence. We have:

$$\begin{aligned} B^{(g+1)} & \leq B^{(g)} - \kappa \sum_{i \in \mathcal{U}} [(\widetilde{x}_{i,j}(t) - x_{i,j}(t))^{(g+1)}]^2 \\ & + \sum_{i \in \mathcal{U}} [(\widetilde{r}_{i,j}(t) - r_{i,j}(t))^{(g+1)}]^2 + \sum_{i \in \mathcal{U}} [\Gamma_{\min} - \tilde{\Gamma}_i(t)^{(g+1)}]^2 \\ & - \kappa \|Y_i^{(g+1)} - Y_i^g\|_2^2. \end{aligned}$$

TABLE II
SIMULATION PARAMETERS

Parameter	Description	Value
σ^2	White Gaussian noise	-174 dBm
$\delta_{i,b}$	Transmission power between the BS b and the UAV i	23 dBm
$\delta_{i,j}$	Transmission power between the UAV i and the vehicle j	23 dBm
L_b	Buffer constraint for tasks transmitted to the edge server	500 MB
L_u	Buffer constraint for tasks transmitted to the vehicle	300 MB
f_b	Edge processing capacity	5×10^7 Hz
f_v	Vehicle processing capacity	2.5×10^6 Hz
$A_{\max}$	Maximal connections for a UAV	3
W_b	Bandwidth between the transmission of the BS b and the UAV i	10 MHz
W_S	Bandwidth between the transmission of the UAV i and the vehicle j	5 MHz
T	Number of large time slots	20
t_a	Duration of large time slots	5
τ	Tiny time interval	0.2 s

Proof: Please get the detailed proof of Theorem 4 through Appendix F, available online. ■

Our strategy can be proven to eventually converge and obtain the optimal solution through the above three theorems.

VI. PERFORMANCE EVALUATION

We consider the UVEC framework in one scenario that has one BS attached to one edge server. UAVs are located in a geographical area whose size is $1\text{km} \times 1\text{km}$. 4 UAVs are deployed in the area following PPP with a communication coverage radius of 250m with unmanned vehicles. Besides, multiple unmanned vehicles are randomly deployed in the scenario and each moves following a Markov jump process model. Each vehicle can connect with three different UAVs. Meanwhile, each UAV can only build up a connection with only one vehicle. The power for transmissions between the UAV i and the BS b and for transmissions between the UAV i and the vehicle j is set to $\delta_{i,b} = \delta_{i,j} = 23\text{dBm}$, and the Gaussian white noise is set as -174dBm . The bandwidth between UAVs and BS is set as 10MHz while the bandwidth between UAVs and unmanned vehicles is set as 5MHz. The length of large time slots and tiny time intervals are 1s and 0.2s. And the amount of large time slots is 20. The processing capacity of the edge server and unmanned vehicles are set as $5 \times 10^7\text{Hz}$ and $2.5 \times 10^6\text{Hz}$. A set of trace-driven experiments are given in this section. The necessary simulation parameters are given in Table II.

A. Methodology

Our JTODAOR algorithm is compared with five different benchmark methods to get a better evaluation of the algorithm. The five compared algorithms are described below.

1) *Edge Server Only Strategy:* Each UAV transmits and offloads tasks only to the edge server.

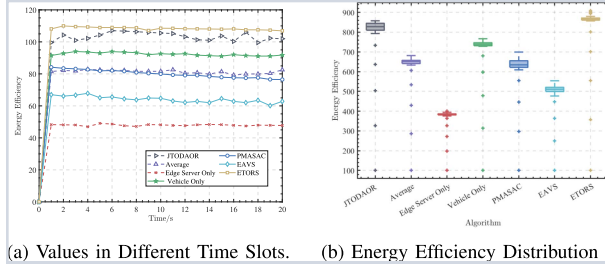


Fig. 5. Energy efficiency analysis.

2) *Vehicle Only Strategy*: Vehicle Only strategy selects the most suitable vehicles to offload tasks for each UAV.

3) *Average Ratio Strategy*: UAVs select vehicles by Algorithm 1 and divide tasks for the edge server and for the selected vehicles equally.

4) *PMASAC Strategy*: The PMASAC strategy is a new parameterized multi-agent RL algorithm based on soft actor critical (SAC) proposed by [26]. The random policy network uses the sum of the negative value of the soft Q function and the logarithm of the current strategy probability as the loss function of the PMASAC strategy.

5) *EAVS Strategy*: EAVS strategy [20] is a deep reinforcement learning (DRL) algorithm based on recent strategy optimization (PPO), which uses a ternary clipping method to make rate decisions to obtain better QoS.

6) *ETORS Strategy*: ETORS strategy [27] is designed to optimize the tradeoff between energy consumption and delay. The offloading strategy is adjusted by Lagrange gradient descent. The optimal value of the Lagrange multiplier related to the original problem is obtained by using the duality technique. In the inner layer, a convex optimization technique is used to solve subproblems effectively.

B. Comparison With Other Algorithms

The value of energy efficiency in each time slot for all strategies is compared in Fig. 5(a). From Fig. 5(a), we can get that the energy efficiency of all strategies fluctuates constantly due to the mobility of vehicles and changes in the task queue. We further observe that our proposed JTODAOR strategy outperforms the other strategies because of JTODAOR's flexible partial offloading mode, which can significantly reduce task processing time. PMASAC and EAVS are dynamic strategies similar to JTODAOR. The PMASAC Strategy is based on SAC, an off-policy algorithm, but the SAC algorithm is easy to diverge in the parallel environment. Hence, the results obtained by PMASAC are worse than the other two strategies. The EAVS Strategy is based on PPO, which is more accessible to converge in a parallel environment than SAC. However, the PPO algorithm is easier to converge to the local optimal, which affects the final results of EAVS. The ETORS strategy aims to optimize the balance between energy efficiency and delay, yet neglects task reliability. Consequently, ETORS achieves higher energy efficiency compared to our algorithm but sacrifices some degree of reliability.

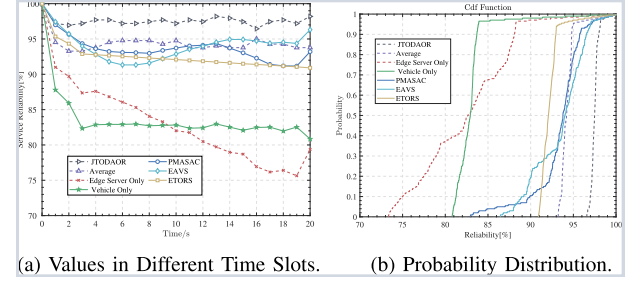


Fig. 6. Total reliability analysis.

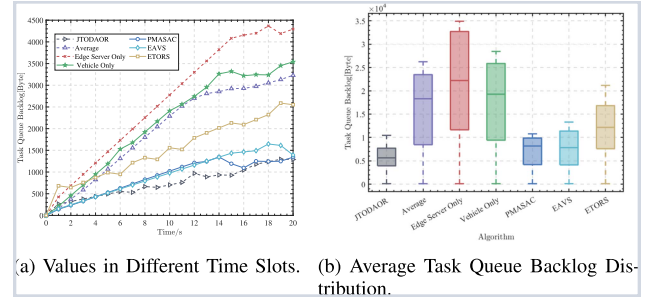


Fig. 7. Average task queue backlog.

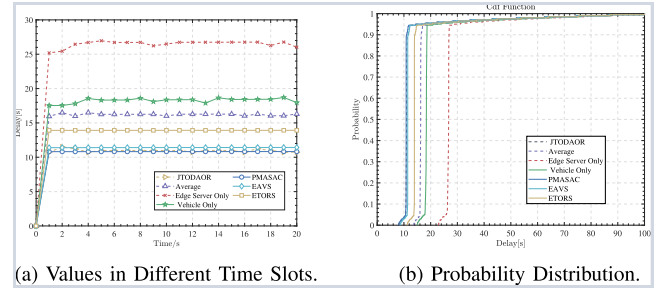


Fig. 8. Average task delay analysis.

Fig. 5(b) is the boxplots of energy efficiency under different strategies, which shows the distribution among different strategies. From Fig. 5(b), it is easy to obtain that our JTODAOR can obtain about 7% higher energy efficiency than the Vehicle Only strategy and about 7.7% higher than the Average strategy and PMASAC strategy, which supports the conclusion in Fig. 5(a). Meanwhile, our algorithm can achieve energy efficiency that is closely comparable to the results of the ETORS algorithm.

Fig. 6(a) shows the total reliability of the six strategies versus the number of time slots. As expected, our proposed JTODAOR strategy outperforms the other schemes, especially the Edge Server Only and Vehicle Only strategies. Moreover, the performance of JTODAOR maintains more stable than other strategies over time, indicating that JTODAOR is more suitable for the changeable distributed scenario. Fig. 6(b) demonstrates the cumulative distribution function (CDF) of total reliability. The Vehicle Only can ensure about 81% of tasks from UAVs experience higher total reliability, while ETORS strategy can ensure 91% total reliability. EAVS and PMASAC can ensure 92%. However,

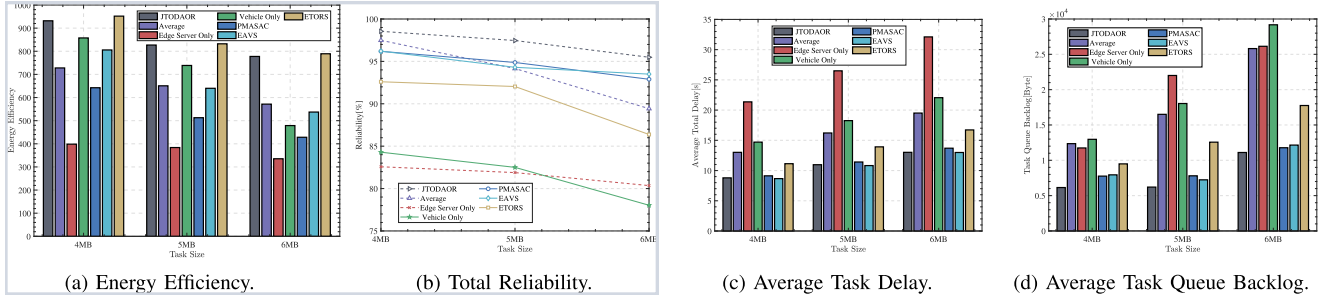


Fig. 9. The impact of task size.

our strategy can ensure 97% of tasks from UAVs experience higher total reliability which outperforms other strategies. Compared to the Vehicle Only strategy, our algorithm also considers offloading tasks to edge servers, which can effectively reduce the impact of vehicle movement on reliability. Additionally, compared with PMASAC strategy and EAVS strategy, our strategy can more accurately predict the impact of vehicle moving trajectory on task queues, so it can achieve better results than these two strategies.

Fig. 7(a) displays the changes in average queue backlog over time slots. The changes in average queue backlog in all strategies rise after a period and then gradually reach a plateau. As expected, our JTODAOR strategy performs better than the other strategies, and the queue backlog changes are the smoothest of all strategies. Our strategy can adaptively adjust the ratio of tasks transmitted to the edge server and the vehicle according to the estimations of queue congestions through SNC. Fig. 7(b) is the boxchart of the average task queue backlog distributions for all strategies. As can be seen from the figure, the fixed task offloading strategies result in a larger queue backlog and greater fluctuation than the dynamic adjustment strategies. Our strategy not only guarantees a minimum queue backlog but also has much less fluctuation in the queue backlog compared to the other strategies.

Fig. 8(a) shows the fluctuation of average delay for tasks a UAV sent at each time slot while Fig. 8(b) demonstrates the probability distributions of average delay for different strategies. From Fig. 8(a), we can find that our strategy JTODAOR achieves the same effect as the PMASAC strategy and the EAVS Strategy but outperforms Vehicle Only and Average strategy and performs far better than Edge Only strategy. PMASAC strategy and EAVS strategy use DRL methods to know about the environment in response to changes in the dynamic environment. Our strategy estimates vehicle mobility and queue backlog to adjust the task offloading ratio dynamically and keep the delay within the required range. In Fig. 8(b), the task delay of the Edge Server Only strategy is about 28s, which is the largest among all policies. Our algorithm can keep the task delay at about 11s, which is almost consistent with EAVS strategy and PMASAC strategy.

Fig. 9(a)–(d) show the simulation results for all strategies under different task sizes. We plot the performance of energy efficiency, total reliability, total task latency, and average queue backlog for various strategies. As can be seen from the figures, the energy efficiency and system reliability of all strategies show a decreasing trend as the task size increases. The average task

delay and average queue backlog length increase with the rise in task size. However, compared with other algorithms, our algorithm always achieves the highest system reliability, as well as the lowest average task delay and average queue backlog length. Although the energy efficiency of our strategy is lower than that of ETORS strategy, the gap between them gradually narrows with the increase of task size. The specific content will be analyzed in the following.

Fig. 9(a) shows the energy efficiency in the six strategies changing with the task sizes. From Fig. 9(a), the energy efficiency decreases with increasing task sizes because a larger task size results in more significant task latency. The larger task latency causes lower energy efficiency and a more considerable queue backlog. Moreover, we can observe that when the task size is 4MB and 5MB, our strategy achieves an average energy efficiency 10% higher than the strategy with the second-highest energy efficiency. When the task size rises to 6MB, our strategy achieves an average energy efficiency of 28% higher.

The total reliability for different strategies changing along with a variety of task sizes is depicted in Fig. 9(b). In Fig. 9(b), the reliability also decreases with increasing task sizes because the larger task sizes result in a more significant delay violation probability. With the increment of task sizes, the reliability of JTODAOR strategy can still be guaranteed at more than 95%, which is better than the performance of other algorithms.

Fig. 9(c) and (d) demonstrate total task latency and task queue average backlog changing under various task sizes for different strategies, respectively. As the task size grows, the total delay and average task queue backlogs are increasing for all strategies. Static algorithms, including Average strategy, Edge Server Only strategy, and Vehicle Only strategy obtain the poorest performance when the task sizes grow. Our strategy always obtains lower task delays and smaller queue backlogs through dynamic resource management and task offloading control. What's more, the fluctuation in our strategy is relatively small when the task size grows.

Fig. 10(a)–(d) show the simulation results for all strategies under different amounts of vehicles. Same as Fig. 9, the performance of energy efficiency, total reliability, average task delay, and average task queue backlog for different strategies are evaluated. As the Edge Server Only strategy will not be affected by the changing of vehicle amount, the curve for the strategy is a horizontal line.

Fig. 10(a) shows the energy efficiency changing with the amount of vehicles among different strategies. As the number of vehicles grows, energy efficiency increases for all strategies.

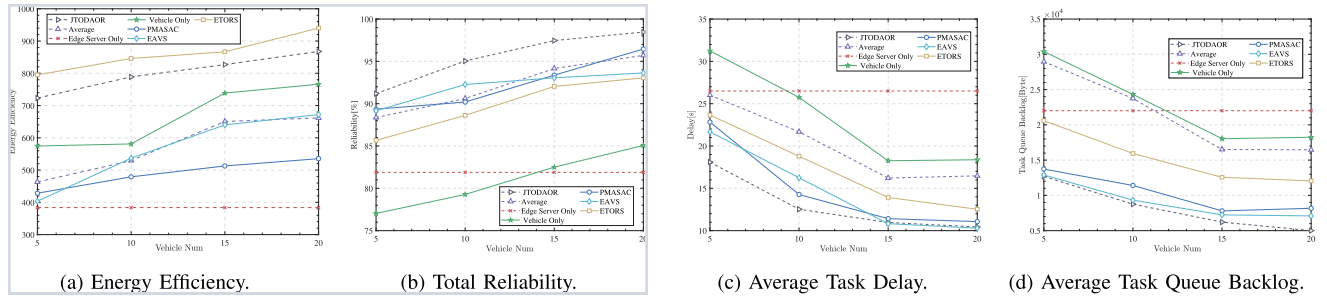


Fig. 10. The impact of vehicle amount.

But from Fig. 10(a) it can be obtained that when the number of vehicles increases from 15 to 20, the energy efficiency change is tiny because the number of vehicles has an upper limit on improving the task delay. For our strategy, increasing the vehicle amount from 5 to 20 can increase the energy efficiency from 700 to nearly 1000. Moreover, the energy efficiency of our strategy is lower than that of ETORS strategy under different vehicle amounts, but the energy efficiency gap between the two strategies is not large. The total reliability, task delay and average task queue backlog of our strategy are higher than that of ETORS strategy, which represents that our strategy can achieve the balance of all attributes.

Fig. 10(b) demonstrates the total reliability for different strategies changing along with different amounts of vehicles. When the number of vehicles rises, the reliability also increases. When the number of vehicles grows, the queue backlogs in the vehicles will significantly decrease, resulting in a more negligible delay violation probability. The increase of vehicle amount from 5 to 20 can increase the reliability of our strategy from 90% to nearly 99%. It is easy to obtain from the figure that our algorithm has an advantage over other algorithms in any number of vehicles.

Fig. 10(c) shows the total task delay changing under different amounts of vehicles for different strategies. Fig. 10(d) demonstrates the average task queue backlog changing under different amounts of vehicles for different strategies. As the task size grows, the total delay and average task queue backlogs significantly decrease for all strategies from 5 to 15. When the vehicle amount rises to 20, the total task delay changes slightly as the vehicle queue backlogs change slightly. Our strategy always obtains lower task delays and smaller queue backlogs under different amount of vehicles.

VII. CONCLUSION

We proposed one novel distributed framework, which is consisted of multiple UAVs in a VEC system. The massive demands for connectivity and the limited resources for UAVs in the framework have brought great difficulties in satisfying the QoS for delay-sensitive services. To maximize energy efficiency while satisfying the statistical delay and queue buffer bounded constraint, we incorporate the optimization framework by integrating vehicle selection and task offloading scheduling. The SNC theory is used to analyze the statistical delay and the maximum throughput with buffer constraints. Besides, we

transform the original problem into a linear problem through a linear transformation method. A consensus ADMM method is proposed to decompose the problem. Through rigorous theoretical analysis and extensive numerical evaluation, we demonstrate the optimality and convergence of our proposed optimization while satisfying service reliability constraints. Simulation results demonstrate significant performance gains over the benchmark approach.

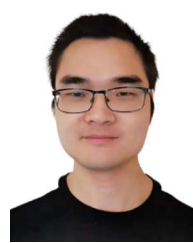
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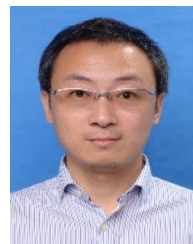


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